INCH-POUND
MIL-DTL-32245
3 February 2023

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, CCU-148/B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the CCU-148/B impulse cartridge, referred to herein as "cartridge", and the methods of verification upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

DOD-STD-2101 - Classification of Characteristics

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SURFACE WARFARE CENTER INDIAN HEAD DIVISION (NSWCIHD) INSTRUCTIONS

NSWCIHDINST 8020.16 - Monitoring of Energetic Materials Containing Nitrocellulose and/or Nitrate Esters

(Copies of this document are available from Naval Surface Warfare Center, Indian Head Division, R Department, ATTN: Deputy Department Head, 101 Strauss Avenue, Indian Head, MD 20640 or online at https://navsea.navy.deps.mil/wc/surinhd/a/private/ac/SitePages/IHD%20Directives.aspx.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

30003 1762AS100 - Cutter, Cartridge Actuated, MLU-57/B

30003 3205AS235 - Cartridge, Impulse, CCU-148/B

30003 3205AS312 - Vibration Test Fixture Assembly, F3

(Copies of these documents are available online via Joint Engineering Data Management Information and Control System (JEDMICS) at https://webjedmics.navair.navy.mil/webjedmics/index.jsp. Refer questions, inquiries, or problems to nani_customerservice@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A276/A276M - Standard Specification for Stainless Steel Bars and Shapes

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Cartridge production. Each new production lot shall pass inspection in accordance with 4.3 prior to acceptance for use.

3.3 Non-destructive evaluation.

3.3.1 Workmanship. Cartridges shall be free of defects (see 4.4.1 and 6.5.1).

3.3.2 Envelope and dimensions. The cartridge shall be manufactured in accordance with NAVAIR Drawing 30003 3205AS235 and all documents listed thereon (see 4.4.1).

3.3.3 Physical configuration. Unless otherwise specified (see 6.2), dimensions shall apply after all machining, process treatments, and performance conditions (see 3.5) have been completed (see 4.4.1).

3.3.4 Primary components. One primary component (see 6.5.4) lot may be used in more than one lot. The propellant used shall be from a lot that has been tested in accordance with and meets the requirements of NSWCIHDINST 8020.16 (see 4.4.1).

3.3.5 Markings. Cartridges shall have proper identification markings in accordance with NAVAIR Drawing 30003 3205AS235 (see 4.4.1).

3.3.6 Radiographic examination. Each cartridge shall be radiographically examined to verify that the explosive charges are properly loaded; that no cracks, voids, foreign material, or discontinuities exist; and that all components are properly assembled (see 4.4.2).

3.3.7 Leakage. The leak rate shall not exceed 1×10^{-5} cubic centimeter per second of air at 1±0.1 atmosphere differential pressure, within 15 minutes of exposure to 27 inches of mercury vacuum for 5 minutes, followed by exposure to helium at 30+2/-0 pounds per square inch absolute for 60 minutes (see 4.4.3).

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3.3.8 Bridge circuit resistance. The bridgewire circuit resistance of the cartridge shall be 1.00±0.10 ohm including connector pins (see 4.4.4).

3.3.9 Power current. The cartridge shall not misfire or degrade (see 6.5.3 and 6.5.2) when subjected to 1 ampere/1 watt direct current (DC) for 5 minutes (see 4.4.5).

3.3.10 Electrostatic discharge. The cartridge shall not misfire or dud when subjected to a 25,000-volt simulated human electrostatic discharge. The cartridge shall meet this requirement at 70±5 °F and a relative humidity (RH) of 50 percent or less (see 4.4.6).

3.3.11 Stray voltage. The cartridge shall be capable of withstanding the effects of a stray voltage environment without misfire when tested in accordance with 4.4.7. The cartridge shall meet this requirement at 70±5 °F.

3.4 Environmental conditions.

3.4.1 Vibration. The cartridge shall exhibit no evidence of cracks, voids, foreign material, or discontinuities and shall meet the conditions of 3.5 when fired after exposure to random vibration testing in accordance with 4.5.1.

3.4.2 Forty-foot drop. The cartridge shall not fire when dropped from a height of 40 feet, +3/-0 inches, onto a 2-inch thick steel plate and shall be safe for handling and disposal after testing (see 4.5.2).

3.4.3 Temperature-shock/humidity/altitude (TSHA) cycling. The cartridge shall meet the conditions of 3.5 when fired after exposure to a 28-day TSHA test (see 4.5.3).

3.5 Performance conditions. The cartridge, when installed in an MLU-57/B cutter, shall completely sever a ¾-inch diameter 6 by 19 independent wire rope core (IWRC) wire rope conforming to 4.6.2 and meet the requirements of 3.5.1 when a 10±0.5-ampere DC pulse is applied to the bridgewire for 1.0 second (see 4.6).

3.5.1 Delay. The time from application of the all-fire current to cable severance shall not exceed 1.00 second (see 4.6.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on cartridges when a first article sample is required (see 3.1). This inspection shall include the tests in the order specified in table I. Failure of any cartridge to meet the requirements of table I shall be cause for rejection of the first article. The first article sample shall consist of 76 cartridges.

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TABLE I. First article inspection.

Test	Requirement	Verification	Sample size						
Visual inspection	3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5	4.4.1	6	12	12	12	12	12	10
Radiographic	3.3.6	4.4.2	6	12	12	12	12	12	10
Leakage	3.3.7	4.4.3	6	12	12	12	12	12	10
Bridge circuit resistance	3.3.8	4.4.4	6	12	12	12	12	12	10
Power current	3.3.9	4.4.5	6	12	12	12	12	12	10
Electrostatic discharge	3.3.10	4.4.6	6	12	12	12	12	12	10
Stray voltage	3.3.11	4.4.7	6	12	12	12	12	12	10
Vibration	3.4.1	4.5.1		12					
40-foot drop	3.4.2	4.5.2	6						
TSHA cycling	3.4.3	4.5.3			12				
Visual inspection	3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5	4.4.1	6	12	12				
Bridge circuit resistance	3.3.8	4.4.4		12	12				
Radiographic	3.3.6	4.4.2		12	12				
Function at:									
-65 °F	3.5	4.6				12			
+70 °F	3.5	4.6		12	12		12		
+160 °F	3.5	4.6						12	
Retain [1/]									10

NOTE:

[1/] Retain units are kept for investigative purposes.

4.3 Conformance inspection. Conformance inspection shall consist of the tests specified in table II. Failure of any cartridge to meet the requirements of table II shall result in rejection of the lot of cartridges.

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TABLE II. Conformance inspection.

Test	Requirement	Verification	Classification (DOD-STD-2101)	Sample/lot [1/]
Visual inspection	3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5	4.4.1	(C1)	Lot
Radiographic	3.3.6	4.4.2	(C2)	Lot
Leakage	3.3.7	4.4.3	(C3)	Lot
Bridge circuit resistance	3.3.8	4.4.4	(C4)	Lot
Power current	3.3.9	4.4.5	(C5)	Sample
Electrostatic discharge	3.3.10	4.4.6	(C6)	Lot
Stray voltage	3.3.11	4.4.7	(C7)	Lot
Function at 70 °F	3.5	4.6	(C8)	Sample

NOTE:
[1/] Inspection performed on test sample or entire lot (including sample).

4.3.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 151 and not greater than 500.

4.3.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table III.

TABLE III. Lot inspection sample size.

Lot size	Sample size	Retained sample
01-50	8	2
51-90	13	2
91-150	20	2
151-280	32	2
281-500	50	2
501-1,200	80	2

4.4 Non-destructive tests.

4.4.1 Examination. Examination of all cartridges shall be made to determine if the cartridge meets the requirements of this specification with respect to workmanship (see 3.3.1), conformance to drawings (see 3.3.2), dimensions (see 3.3.3), primary components (see 3.3.4), and markings (see 3.3.5).

4.4.2 Radiographic examination. Each cartridge shall be radiographically examined at 90 degrees to the longitudinal axis in accordance with ASTM E1742/E1742M. Any observable imperfections (i.e., defects) in the assembly shall be cause for rejection. The cartridge shall be positioned for the most revealing exposure with the long axis of the cartridge perpendicular to the radiation from the X-ray camera. All cartridges shall be identified with a serial number prior to examination. The cartridge serial numbers shall be consecutive beginning with the lowest number in each production lot. The cartridges shall be arranged in trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridge's display thereon. Discontinuities in serial numbers shall be identified on the radiograph and those defective cartridges removed from the lot. Each radiograph shall display the following information:

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a. Nomenclature.

b. Department of Defense Identification Code.

c. Part Number.

d. Lot Number (in accordance with MIL-STD-1168).

e. Serial Number (in numerical order from lowest to highest).

f. Contract Number.

4.4.3 Leakage. A leak test shall be conducted with a dry gas leak detector of sufficient sensitivity to ascertain if cartridges meet the leak rate requirement of 3.3.7.

4.4.4 Bridge circuit resistance. The resistance of each bridge circuit shall be measured with an accuracy of not more than 1 percent using a test that subjects the bridge circuit to a current of less than 25 milliamperes. The measurement shall be made at or corrected to 70 °F.

4.4.5 Power current. To determine if the cartridge meets the power current stimulus of 3.3.9, a direct current of not less than 1 ampere supplying a minimum of 1 watt shall be applied to each bridge circuit for a period of at least 5 minutes. The current shall be applied to all bridge circuits. The test current shall be regulated throughout the period of application to within 2 percent. If a rectified current is used, the ripple content shall not exceed 5 percent root mean square of the test current. The cartridges shall be conditioned at 70±5 °F or 200±5 °F, as required in table I, for a period of 12 hours.

4.4.6 Electrostatic discharge. To determine if the cartridge meets the requirements of 3.3.10, a 500±5-percent picofarad capacitor charged to 25,000±500 volts and a resistor shall be connected in a 5-microhenry total inductance series circuit between pairs of pins or leads in all combinations and between the pins and the case of the initiator. The test shall be performed using both a 5,000±5-percent ohm and with a 500±5-percent ohm resistor. The series connection shall be maintained for 60 seconds. Switching in this circuit shall be accomplished by bringing together two 0.5±0.05-inch spherical metal electrodes from an initial separation of 3 inches in air. Each series test shall constitute a separate test. Cartridges used for this test shall be temperature conditioned for a minimum time of 12 hours at 70±5 °F.

4.4.7 Stray voltage. To determine if the cartridge meets the requirements of 3.3.11, each cartridge shall be subjected to 2,000 pulses of DC. Each pulse shall be 300 milliseconds in duration and pulse rate shall be 2 per second. Each pulse shall have a minimum amplitude of 100±5 milliamperes. Cartridges used for this test shall be temperature conditioned for a minimum time of 12 hours at 70±5 °F.

4.5 Environmental tests.

4.5.1 Vibration. The vibration testing shall be conducted using a fixture that conforms to vibration test fixture assembly F3 in accordance with NAVAIR Drawing 30003 3205AS312. The test shall be conducted with all 12 cartridges installed in the test fixture. The testing shall be 1 hour in the longitudinal axis per temperature and 2 hours in the transverse axis per temperature. All 12 cartridges shall be subjected to all three temperatures, -65 °F, +70 °F, and +200 °F. The vibration test shall be conducted as follows:

a. The cartridges shall be subjected to fixed dwells at 70 °F to the levels specified in table IV along the longitudinal axis.

TABLE IV. Fixed dwells along longitudinal axis.

Frequency (hertz)	Acceleration (G-force)	Duration (minutes)
20.9	3.125	7.5
146.3	2.25	7.5
292.6	2.25	7.5
438.9	2.25	7.5

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b. After fixed dwells along the longitudinal axis, the cartridges shall be subjected to vibration along the longitudinal axis for 1 hour with a 7.5-minute sweep to the vibration levels shown on figure 1.

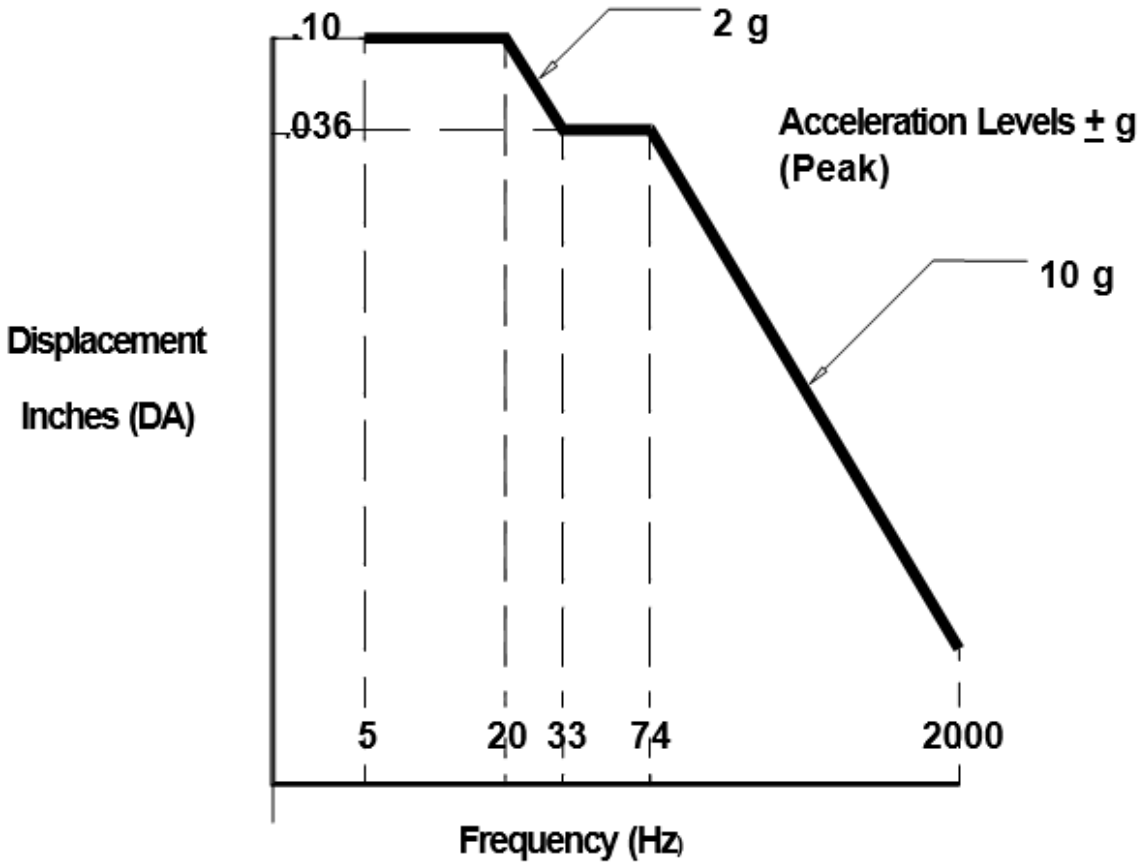


FIGURE 1. Vibration test level.

c. After completion of vibration along the longitudinal axis, the cartridges shall be subjected to fixed dwells at 70 °F to the levels specified in table V along the transverse axis.

TABLE V. Fixed dwells along transverse axis.

Frequency (hertz)	Acceleration (G-force)	Duration (minutes)
20.9	3.125	1.5
146.3	2.25	1.5
292.6	2.25	1.5
438.9	2.25	1.5

d. After fixed dwells along the transverse axis, the cartridges shall be subjected to vibration along the transverse axis for 2 hours with a 7.5-minute sweep to the vibration levels shown on figure 1.

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e. After completion of vibration at +70 °F, the vibration procedure shall be repeated at +200 °F. After completion of vibration at +200 °F, the vibration procedure shall be repeated at -65 °F.

4.5.2 Forty-foot drop. Six unprotected cartridges shall be dropped and impacted onto a 2-inch-thick steel plate supported by reinforced concrete, from a height of 40 feet, +3/-0 inches, in each of the following orientations: two unprotected cartridges nose up, two unprotected cartridges nose down, two unprotected cartridges horizontal. The plate shall have a flat surface (not deformed from previous test impacts to the point where further proper angular impacts are prevented) having a Brinell hardness of not less than 200 HB and shall have a length and width of at least 2 times the maximum dimension of the test item. The cartridges shall be guided to impact these surfaces in the proper orientation. Surface damage from impact is allowed.

4.5.3 TSHA cycling. The cartridges shall be supported on screen trays so that all areas are exposed to the prescribed atmospheric conditions at all times. The cycling scheduling shall be as follows:

Monday	0800	Place test items in a chamber maintained at +70 °F at 50 percent RH.
	1200	Raise chamber temperature to +160 °F and the RH to 95 percent. The chamber temperature shall reach +160 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 psi).
Tuesday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70 °F at 50 percent RH.
	1200	Remove test items from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above and immediately place in a chamber maintained at +160 °F at 95 percent RH.
Wednesday	0800	Reduce chamber temperature to +70 °F at 50 percent RH. The chamber temperature shall reach +70 °F at 50 percent RH not later than 0900.
	1200	Raise chamber temperature to +160 °F at 95 percent RH. The chamber temperature shall reach +160 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 psi).
Thursday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70 °F at 50 percent RH.
	1200	Remove test items from the above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above chamber and immediately place in a chamber maintained at +160 °F at 95 percent RH.
Friday	0800	Reduce chamber temperature to +70 °F at 50 percent RH. The chamber temperature shall reach +70 °F at 50 percent RH not later than 1300.
	1200	Raise chamber temperature to +160 °F at 95 percent RH. The chamber temperature shall reach +160 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65 °F at standard ambient pressure.

This schedule shall be followed for a total of 4 weeks (28 days) except that, on the second and fourth weekends, the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +160 °F at 95 percent RH.

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4.6 Ballistic performance. The cartridges shall be conditioned not less than 4 hours at 70±5 °F. First article test units shall be conditioned not less than 4 hours at -65±5 °F, +70±5 °F, and +160±5 °F in accordance with table I. Cartridges shall be installed in an MLU-57/B cutter (in accordance with NAVAIR Drawing 30003 1762AS100) and fired in a horizontal position against a ¾-inch diameter 6 by 19 IWRC wire rope in accordance with 4.6.2, preloaded with a 200±10-pound tensile load.

4.6.1 Delay. The time from application of the all-fire current to cable severance shall not exceed 1.00 second.

4.6.2 6 by 19 Warrington wire rope.

4.6.2.1 Construction. All outer strands shall be Warrington construction. The IWRC shall be of 7 by 7 construction. Preforming of the outer strands shall be required. The wire rope may be right hand or left hand lay. The strands of wire rope shall contain no joint made after fabrication. If joints are made in individual wires before fabrication of the strands, the distance between a joint in one wire and a joint in any other wire in the same strand shall be not less than 18 inches. The wire rope shall conform to table VI.

TABLE VI. Requirements for 6 by 19 Warrington wire rope.

Nominal and minimum diameter	Maximum diameter	Approximate weight	Minimum breaking strength	Rope pitch minimum	Rope pitch maximum
¾ inch	0.781 inch	1.05 pounds per foot	45,700 pounds	4.900	5.438

4.6.2.2 Material. All strands shall be fabricated from Unified Numbering System alloy S30500 steel wire in accordance with ASTM A276/A276M. The wires shall have a minimum tensile strength of 210,000 pounds per square inch. All material incorporated in the wire rope shall be new and may be fabricated using materials produced from recovered materials (see 6.5.5) to the maximum extent practicable. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridge was developed for use in the MLU-57/B cable cutter. The MLU-57/B cutter is only used in military aircraft; therefore, the cartridge has no commercial application. Current stock of the CCU-148/B cartridge assembly may be used until depleted; future requisitions for the cartridge should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1).

c. Dimensions, if other than specified (see 3.3.3).

d. Lot size, if other than specified (see 4.3.1).

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e. Sampling, if other than specified in table III (see 4.3.2).

f. Packaging requirements (see 5.1).

g. The facility approved for first article inspections (see 6.3).

h. Data standard deliverables, when required (see 6.4).

6.3 First article inspection. All parts and materials, equipment, and processes and procedures, including packaging and packing, must be the same as used for regular production and must be obtained from the intended sources of supply for production at the facility specified (see 6.2). A first article inspection will be required if 2 years or more have passed since the cartridge or a similar item was manufactured, major turnover in personnel has occurred, major changes to the production facility have occurred, or production has transferred to a new facility or site.

6.4 Standard deliverables. In addition to the delivery of cartridges, the contract may require the delivery of the following:

a. Qualification test plan.

b. Qualification test report.

c. Radiographic sample plates and technique.

d. Hazard classification plan.

e. Requests for waiver/deviation and engineering change proposal.

f. Hazard classification report.

g. Insensitive munition test report.

h. Ammunition data card.

i. Process control charts.

6.5 Definitions.

6.5.1 Defect. For the purposes of inspection or verification, the term defect is defined to mean burrs, dents, scratches, split or cracked edges, sharp edges, and all other defects that may prevent installation, adversely affect the ballistic performance, or in any way adversely affect fit, form, function, safety, or performance of the cartridge.

6.5.2 Degrade. For the purposes of inspection or verification, the term degrade is defined to mean chemical decomposition, fracture, yielding, debonding, delamination, cracking, or any other condition that could adversely affect the form, fit, function, or safety of this component.

6.5.3 Misfire. A misfire is defined as failure to meet any performance condition.

6.5.4 Primary component. Primary components are defined as any explosive, propellant, delay composition, or pyrotechnic material.

6.5.5 Recovered materials. The term recovered materials means materials that have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials.

6.6 Subject term (key word) listing.

MLU-57/B

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – OS
Air Force – 11

Review activities:
Army – AV, TE
Navy – AS
Air Force – 84, 170

Civil agency:
GSA – FAS

Preparing activity:
Navy – OS
(Project 1377-2021-020)

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